

ActInf Textbook Group ~ Cohort 4 ~ Meeting 12 (Chapter 5 part 2)

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All right, welcome. Thanks everyone for joining. This is the final recorded session on the first half of Cohort 4. It's August 22nd, 2023, and we are going to discuss Chapter 5.

So before we look at what we might have ahead of us from questions or anything else, does anyone want to make any general comments on 5?

I'm just, anyone can give a thought.

But just since we're on Chapter 5 in the neurobiology, we talked, I think, a little bit in the Chapter 10 closing discussion for Cohort 3 about some of the topics like what is empirical evidence or what does any experimental setting have to do with free energy principle active inference.

And we reflect a little bit on this recent paper that, like, with Isomura et al, that, like, broke through kind of another level, not necessarily even from an epistemic perspective.

Oh, I guess in terms of people's awareness, it was epistemic in that sense.

Then I was thinking about this was in 2018.

I mean, it's the same title.

I know that the work has proceeded a lot since then, so that's not to say they rehashed the same.

But in some ways, because there is a difference between map and territory, between, like, models of X and X, in a way, there's, like, this eternal gap that just continues to swallow up experimental results, much to the chagrin of people who think that just one more experiment is going to bridge or map the gap.

And that's kind of, like, what some of us are told as graduate students.

A lot of graduate school has to do with, like, creating a data set, and people get into the zone of thinking that that one data set is going to be addressing a question, even a narrow one.

And of course, like, if it's not helpful, then let's not waste our resources to do that.

But at the same time, people were saying that the relationship between gene expression and behavior was resolved in the 90s.

And it's the same language that comes out today.

And then even in this kind of fractal free energy way.

It's like, what were we looking for in the Petri dish again?

Oli?

Yeah, that was exactly what I was going to say.

Because, in fact, I would love to see in newer editions of the textbook to see some of the new directions in what some people call radical biology to be incorporated much more than we see currently in the textbook.

Because, actually, the works of scientists such as Michael Levin and even before that, I mean, Varela, Maturana, Maturana, and Kapra, they show some radically new approaches to see biology and even life itself through, I mean, the conceptions of autopilot and allogenesis and other related terms.

But I believe that closer incorporation of these concepts into the textbook can help to see the bigger picture, which connects Bayesian mechanics as an overarching theory of every, quote-unquote, thing to active inference.

Because the way it stands now, and in particular, the way Chapter 5 stands now, there seems quite a big gap between what Bayesian mechanics represents as a theory of everything and its neurobiological representation or, I don't know, neurobiological evidence or whatever.

But if there was, I mean, more material on these new directions of radical biology, I believe these gaps can begin to be filled and the bigger picture can be, can finally emerge as a kind of continuous picture with, without any significant and possibly confusing gaps in between.

Okay.

Anyone else want to give a thought on that?

Just one speculation here.

The map territory, theory practice, the sort of dualistic dialectic is well-known.

It's about questions like, what does math have to do with biology?

And what does statistics have to do with something like a system of interest?

So that's kind of the classic.

We're dealing with another theoretical distinction.

Within the domain of theory, we're making a distinction, not a homogenization, but a distinction between the more theoretical physics axiomatic view and the lens or tradition or method or approach or process theory described by active inference.

That level of disentanglement of kind of like the platonic core from the more manifest models in theory, but theory in practice.

And that's like exactly where active sits.

So I think in the live chat for the symposium session, David and I were kind of going back and forth.

And his contention was like, well, active inference has been like really useful in the machine learning and artificial intelligence space, but it hasn't, it hasn't made any contributions to neuroscience.

And then, I mean, there's so many angles to take on that point, but like one is, is that it hasn't made many contributions relative to what you think another theory has done?

Or is it, it hasn't made enough contributions relative to what your preferences for what it could have done by this time are?

That's one angle.

But then the other angle that kind of presented itself was like, like, are we looking for experimental validation of the free energy principle?

Or are we looking for people who want to apply active inference?

And that's the experimental validation.

I mean, what if it was experimental validation of some of squares linear regression on electronic healthcare record data?

Be like, talk about a non sequitur.

Decade after decade, experimental validation of linear regression on rainfall and tree growth data.

How many decades before you just say we used a linear regression?

And if you think we could have used it better, then tell us how.

Or if you have another model other than a regression, let's see it.

But that mobilizes the discussion in like a very radical way.

Ali?

Yes.

And I also don't think that active inference or FEP can be considered as a kind of niche theoretical thinking anymore.

Because recently, more and more often I come across some articles or books that don't mention active inference or FEP at all.

Or I don't know any work related to active inference.

But the ideas presented in these works are very, very similar to what active inference, to active inference as assertions.

So it looks like there's the box and active inference or FEP fits right into it.

I mean, it's a theory that is proposed, I mean, at the right time and for the right audience, I believe.

Because if it were proposed, it was proposed about, I don't know, three decades ago, maybe or probably it could go into oblivion without people being aware of any of its significance at all.

But as I said, it seems like the present time is kind of fertile for these kinds of not radical, but a bit out of the box thinking that can potentially help to reshape our understanding of life in its broader sense.

So, yeah, again, its current status seems to me like it's beginning to be mainstream or at least the early phases of mainstream theory is truly evident from some recent work in this area.

Yeah.

Yeah.

Yeah.

And it's so fun to see the memes and the narrative sharpen.

Like when I was just walking around a few minutes ago, I was thinking, okay, reward maximization or precision optimization on getting more reward.

Same thing.

But also surprise bounding is more general than reward.

But you could make a pure reward model.

We see that in the textbook.

So it's like, what's your first principle?

Reward?

Surprise bounding?

Or something else?

Or nothing?

Or you haven't thought about it?

It's all fine.

But which one of them is it?

Or do you want to pop out a level and even have like a different way of approaching the question?

So it's not like presenting a closed box.

It's like a more dynamical frame, which is largely, potentially even overwhelmingly congruent with simply modern statistical physics.

And again, then do you want to say, well, I have an alternate statistical physics branch?

Or I have higher visions for what statistical physics could do than what it is doing?

It's like, congratulations.

Then you're a researcher.

Now, it's still maybe decades or never until you make a contribution.

But that's exactly the path of a researcher.

Seeing an area and then thinking that it can go beyond.

And like, well, I don't have that alternative proposal for niche construction yet.

But I want to be in the game to be there to recognize when it does happen.

Or I want to make something happen eventually.

But merely waiting at the starting line, comparing frameworks to undescribed alternatives.

We barely even have the vocabulary to describe some of these conversational patterns.

And now we have more flexibility to like pop out and pull back to the meta science.

But again, it's like, okay, so are you saying we can't use math to use to do in the context of neuroscience?

And some people may hold that perspective.

And that may be the conversation to have at that time.

But once the, or somebody may think that the engineering tools, like Bert brought this up many times in the symposium.

That machine learning benefits greatly from decades of development on toolboxes.

And in his section, he was like, yeah, this is like the optimistic, positive message is within months to years.

And Carl also alluded to it in the roundtable.

Like, we will have closer to push button solutions.

So I think that really contextualizes a lot of what we're doing and learning.

When a lot of these formalities are going to get just like simply baked in.

And people will be engaging at more applied layers.

But all in the context of chapter five, where a variety of empirical data are being brought into play.

We're talking about real anatomy.

Talking about real reflexes, neurotransmitters, real design patterns in the nervous system, in the body.

So does anyone want to give a thought or ask any chapter five related question?

All right, let's just see if we can upload, just see what comes through.

Okay.

Schematic.

In figure five, one shows layer five of the cortex projects to spinal pyramidal neurons.

This can be interpreted as a prediction.

Spinal motor neuron and spinal pyramidal neurons are sometimes used interchangeably.

Sp.

Okay.

Anyone have a preference for which one of these questions to jump into?

Sp.

I would be curious about the question regarding how do habits and goals interact?

I found this chapter interesting in part just because the introduction of habits are like the

E variable, I suppose.

Yeah, well, let's start with what is the goal.

We're going to have to reach outside of the active inference ontology to bring it in because we don't need goals.

Although at the same time, the term is used prolificately.

Let's see how it's used in five.

This is a mixed, mixed dialect.

E and G are habitual and goal-directed drives.

That's a very interesting sense of goal.

Like in the football, soccer setting, the goal is like a fixed external objective, like a milestone, but a culminating milestone.

But how is it being used here if by goal we mean something that G sharpens?

Well, in that case, well, first let's look at what E does.

E is the prior on policies.

So it reflects before the observation comes in, before updating has happened.

What are the prior probabilities of certain actions being taken?

And then that observation may not influence E.

So E kind of gets passed through, and we see that in the basal ganglia, second example in the chapter, or it gets sharpened.

So if we see expected free energy minimization as being goal-directed, certainly it's not sufficient to say that goals are pragmatic only.

Because if we take everything that G encompasses, like that better policies as evaluated by G are more goal-directed, and inferior policies are less goal-directed.

Then, up in the category of more goal-aligned, like let's just say the two most goal-directed policies, they may radically differ in their composition because one may be heavily loaded on pragmatic value, homeostasis, bringing observations into alignment with preferences and all that.

And the other policy might have a good expected free energy because of its high epistemic value.

And so it seems like at least just locally here, though probably elsewhere in the book they're talking about

goals in just this kind of broader cybernetic goal-oriented sense.

Here we see the goal is like the process, and the goal is this imperative.

Or maybe there's other ways to read it too, but it's kind of like G is - it's like if you were trying to drive somewhere, there's like a location.

You're trying to get there with a location and a time.

You could also think about that as precision in space and time.

Which decisions are going to be more goal-oriented?

Well, if the goal is just the location on the soccer pitch, then the most goal-oriented policy is the one that's like locally or globally taking you closest to the location on the soccer field.

So that's a situation where you're like, yeah, there's time, but really like let's just pursue the space or location as fast as possible.

We'll get there when we get there.

But then if you say, well, we're going to hold up both of these features in this driving example, the space and the time, but in G, the epistemic and pragmatic, and we're going to say, well, actually the joint optimization of those two is what we're going to call our goal.

That's a little different sense of goal.

But it is used here, and it's specifically used to talk about how habits and goals differ, namely where habits are reflecting the prior probability on taking different actions, kind of the pass-through model, where observations don't influence what actions are likely to be taken.

Though still actions may be taken at unequal frequencies, whereas goal orientation means an increasing role for expected free energy in sharpening that policy prior into policy posterior.

Ali?

Oh, sorry.

Well, yeah, another relevant remark here is about this.

You see, there's this kind of telephobia among biologists, and they usually try to avoid using the word purpose instead of goal,

because they believe that although biological agency implies a kind of goal-directed autonomy,

it doesn't necessarily require for those agents to have any kind of purposiveness, as Dennett would argue.

But the problem here, I mean, the problem here, I mean, the distinguishing between these two concepts is a kind of, I mean, as I said,

it probably emerges from this telephobia among biologists, because any talk of purposiveness can be seen as provoking teleology,

which is something that scientific culture has been very averse from many centuries ago.

But if we somehow define goals or purposes as a kind of mechanistic lawfulness to apparently purposive behavior,

then this dilemma between these two concepts would probably be resolved,

because in this case, we have basically defined a goal in terms of purposiveness.

So they're not actually separate terms, and they co-constitute each other.

Of course, the same kind of phobia was evident probably much more,

it was a bit more prominent a couple of decades ago,

but it was also evident for the concept of function.

But it seems like this phobia has been diminished in recent years,
but nevertheless, this telephobic refusal to accept purposes as real factors in biology still remains.
But the active inference and some of these recent progresses in biology,
as I said, the works of Michael Levin and others,
are ways to somehow dismantle this dichotomy and this telephobia,
and as David Krakauer would call,
to see complex systems as a kind of teleonomic matter.

So it doesn't necessarily be something that would require us to think about any,
I don't know, supernatural or supernatural entities or transcendental entities
to get into the picture.

We can talk about purely naturalistic and mechanistic teleology as well,
using these goals and purposes in this kind of interrelated terms.
Yeah, great comments.

One more piece on that from like the so-called argument by design,
which has actually played a very interesting role arising,
most famously in the watchmaker analogy,
which is like you're walking on,
putting aside the big meta,
which is like what is the utility of any thought experiment
or any like little narrative walkway that somebody provides you?

It just is like if you look at a watch,
you're going to infer intention of a creator.

And so then the argument sometimes plays out like,
well, look at all the small bones in the ear
or look at the way that the retina is.

Then the design argument became actually one of the design and engineering
became some of the key pieces of the modern evolutionary synthesis
in evolutionary biology.

But as Ali alludes to,
there always was this kind of teleophobia,
as Mike Levin calls it sometimes,
or just like this awkwardness around like,
well, are we going to appeal to just all functions just as they are?

So that's like a sort of like functional teleology,
just or does one go the kind of Teilhard de Chardin,
Omega Point, Bobby Azarian angle?

Like that is the account.

And where the radical biology in a way speaks to that
or cuts the knot is like the hidden fallacy in the watchmaker fuss
is like it tries to transfer the analogy of a top-down function bestowment.
Whereas if we have self-creating or co-creating or embedded arisal,
then we can say,
yeah, like the circadian circuit
arose through evolution and development
as a bottom-up self-created in relation
mechanism for just that telling time.
So at the level of linguistic core straining,
it makes sense to say that the circadian clock keeps time,
just speaking super broadly here,
like a watch keeps time.
But there's a critical difference,
which is that the watch was like an engineering artifact
as the top-down projection of a cognitive system,
whereas some kind of like internal organic representational system,
like a circadian clock,
is actually like a bottom-up construction.
So like there just isn't an issue
with describing or understanding
goals.
And it's just an English word anyway.
Once we really start to get into the specifics
with agency and habit,
a lot of this
may just dissipate.
And in the more recent multi-scale formalizations,
the habits
can include like
the phylogenetic scale.
Or the habit could refer to just
what is passed through moment-to-moment
in the organism's life.
Or you can imagine this as being like
a intergenerational transfer.
And a habit or a developmental tendency
for a habit is also

able to be inherited.

Andrew, any thoughts on that?

Or wherever anyone else wants to go?

No, I don't have any

any deeper thoughts on that.

I suppose.

I guess I'm more interested in the

just sort of computational aspect of it

as far as message passing goes rather than

I mean, I find all of the

kind of the more abstract philosophical

side of discussing what goals are

very interesting and in certain ways

very important

when putting together a framework.

But I guess here it's just

I just found it interesting like

oh, now we have habits

in addition to

G

which I

previously

considered to be like

just

expected free energy.

But maybe I have my terminology

mixed up here.

Is this G

not the same as

Oh, you're totally right.

And I think

that's just

it's a little

coincidental

alliteration.

and

G

never stands for goals.

It's just a total
typographical
spandrel.

So then the question becomes
what is the goal concept
broad enough to encompass
what it is that G
does?

and that's
clearly not the same thing
as just like
the mile marker
on the side of the road.

But then a little bit more
on the
computational side

E
primes the pump
for the procedure
that G
does.

like the shape
of the variable

E
is the same
shape
that gets taken
into

G.

The cells
in the
in the
variable

E
are
what get
sharpened
by G

as it
iterates
through every
policy
and
calculates
its expected
free energy.

So E
and G
they both
are engaging
with this
policy variable
which is just
a list
of all the
possible policies
so all the
affordances
basically
raised to the
power of how
many time
steps that you
want to
play it out
for
and
 G
sharpens
what E
provides
or modifies
what E
provides
and the
gamma

the temperature
parameter
on
decision
making
here
associated
with
dopamine
neuromodulation
is
modulating
that trade
off
between
the
E
pass
forward
in the
indirect
pathway
there's
kind of
type one
policy
and then
policy
updating
driven
by G
sure
no
thanks for
kind of
giving more
explanation
on that

it's
so I
find that
very
interesting
then
because
at some
point
in here
it refers
to
like
sorry
I can't
find it
immediately
but the
use of
the word
context
so
it's as
if
E
so those
are your
priors
over
policies
and so
you're
lacking
something
like
context
of the
given

the agent
is lacking
context
in a given
moment
I suppose
such that
one might
like
rely
upon
their
habits
to
deal
with
the
scenario
whereas
G
is in
fact
taking
into
account
not
only
E
but
context
in the
sense
of
oh
the
agent
is
taking

in
observations
it's
computing
expected
free
energy
I guess
I'm just
putting
everything
together
as far
as I
understand
it
so
it's
E
prime
you
start
with
E
you
enter
into
some
context
with
a
variety
of
observations
or
otherwise
from
there

you
begin
to
compute
G
and
then
from
there
you
can
start
to
actually
score
potential
policies
to be
taken
with
some
of
them
deemed
to be
more
minimizing
free
energy
given
the
context
and
so
on
versus
E
which

seems
to be
more or
less
static
through
the
whole
computation
does
all
of
that
track
yeah
great
points
so
definitely
when you
set it
up
you
could
make
it
so
that
E
is
fixed
or
that
you
can
make
it
so

that
E
counts
like
earns
that's
one
simple
learning
technique
but
it's
a great
connection
with
context
so
let's
say
that
we
are
playing
cards
and
it's
like
the
main
idea
is
we
have
beliefs
about
how
we
act

in
particular
context
you
know
I
we're
playing
21
blackjack
and
I
the
going
to
sharpen
through
G
again
that
doesn't
even
mean
it's
better
than
playing
on
reflex
it
doesn't
mean
it's
better
than
playing
50
50

it
doesn't
mean
you're
going
to
win
it
doesn't
mean
like
all
these
like
kind
of
outcome
oriented
possibilities
in the
world
but
it
is
the
case
like
if
one
lets
the
card
that
they
saw
influence
their
decision

making
we
would
model
that
as
the
generative
model
using
expected
free
energy
based
policy
evaluation
planning
as
inference
even
if it's
just
one
time
step
planning
that's
how we
understand
the
situation
where
a person
who
stays
80%
of the
time

looks
at the
cards
and
makes
a
decision
that
differs
from
that
or
it
could
be
who
else
is
at
the
table
and
then
so
then
they
want
to
make
a
certain
move
not
because
it
aligns
their
pragmatic

value
with
winning
the
pot
but
that
it
gains
them
information
about
who
they're
playing
with
and
their
strategies
which
speaks
to
this
kind
of
thinking
through
other
minds
topic
and
also
just
the
need
to
build
specific

generative
models
because
how
can
we
talk
about
what
is
an
expected
free
energy
minimizing
!
decision
in the
context
of a
blackjack
game
unless
you've
delineated
what
is even
being
considered
but
now
think
about
how
far
people
go
beyond

the
blackjack
game
talking
about
what
does
and
doesn't
reduce
expected
free
energy
no
thanks
again
for
that
that
was
very
clarifying
and
yeah
so
suddenly
the
phrase
is
model
free
versus
model
based
systems
in this
context
makes

a lot
more
sense
yeah
model
free
just
kind
of
flailing
but
that
may be
an
effective
flail
and
it
may
be
an
effective
way
to
go
or
model
based
is
where
you
get
more
explicit
structuring
okay
this
one's

a
freebie
popcorn
in the
chat
or
however
what
other
neural
systems
are
interesting
to
model
other
than
the
three
that
are
in
chapter
five
what's
another
mammalian
neural
region
or
non
mammalian
system
that
would
be
fun
to

model
actually
I
would
very
love
to
see
active
inference
applied
for
modeling
emotion
circuitry
and
in fact
it's an
ongoing
project
of mine
but
it proves
much more
complicated
than I
previously
imagined
but
well
some
preliminary
works
have been
done
by
people
like

Ryan
Smith
others
but
there
still
needs
lots
of
work
for
this
kind
of
modeling
to
be as
mature
and
sophisticated
as
some other
areas
that active
inferences
has been
applied to
but
for me
personally
it's
really
interesting
prospect
to see
how
modeling
emotions

are
could
train
and even
modeling
emotion
itself
using
active
inference
terminology
and
concepts
would reveal
some
deep
mysteries
about
emotion
and
its
relation
to
cognition
and
other
mental
faculties
things
so
that's
one
area
of
interest
cool
a few
things

that
Dr.
Anna Lemke
said today
in the
roundtable
that
stuck with
me
one
was
about
this
co-presentation
I won't
even call
it
a tension
or a
paradox
between
accepting
higher
power
and taking
agency
in one's
own
life
which
she
related
to her
context
in the
psychiatry
and addiction
setting

and then
also
it came
up
later
with
precision
and
monitoring
and the
increasing
ability
for
monitoring
of
our
biometrics
and
just the
bigger
picture
on how
maybe
knowing
your
heart
rate
at all
times
maybe
it spirals
off in
all these
different
ways
like
now you
have too

much
precision
on
heart
rate
you're
paying
too
much
attention
to
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or
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just a
lot
of
interesting
pieces
there
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about
limitations
of the
brain
as a
predictive
system
or
of
modeling
the

brain
as a
predictive
system
and
yet
again
another
place
where
action
comes
in
to
save
the day
slash
slam
dunk
because
it's
like
when
it's
a
prediction
about
what
you're
about
to
do
it's
!
it's
only
a
prediction

for
so
long
and
then
it
happens
so
it's
like
a lot
of
these
like
prediction
like
they're
often
implicitly
in
this
kind
of
like
archaic
dark
room
setting
like
where
actions
can't
be
taken
and
it's
just
a

perceptual
question
but
we know
that
action
plays
a role
in
perception
itself
through
the
eye
saccade
and
through
the
example
at the
beginning
of the
textbook
where
you
can't
determine
the
texture
of
a
fabric
without
moving
your
finger
so
it's

like
well
it's
not
demeaning
to say
that
it's
predictive
when
it
gets
implemented
a
split
second
later
but
so
much
of
the
predictive
discussion
focuses
just
on
predictive
analytics
predictive
forecasting
what's
going to
be the
most
popular
song
next

year
and
that's
a
very
removed
way
of
considering
the
role
of
action
and
inference
but
yeah
a lot
of
neural
systems
to
model
okay
just
kind
of
a
related
question
what
point
does
oh
I'll
go
for
it

yeah
I
just
wanted
to
make
a
maybe
a
side
way
remarks
but
there
is
even
this
recent
trend
toward
the
probabilistic
aesthetic
theory
of
art
so
especially
as
applied
to
avant-garde
or
contemporary
art
so
and
they

call
it
predictive
arts
so
one
pioneer
in this
area
is
Konrad
Wojnowski
who
also
has
an
upcoming
book
I
think
it
will
be
published
in
February
2024
but
yeah
as you
said
these
kinds
of
probabilistic
modes
of
sensing

and
making
sense
of
reality
has
started
to
being
applied
to
even
aesthetics
and
I
mean
contemporary
art
criticism
as
well
well
yeah
I
remember
this
this
paper
this
was
I
was
in
the
the
the
buy
it

based
on
the
title
phase
of
my
graduate
school
access
to
free
printing
and
I
just
remember
thinking
this
was
like
the
funniest
and
boldest
title
because
it's
like
several
sentences
and
it
has
all
these
weird
illusions

!
but
on
the
left
side
here
is
the
time
scales
like
from
the
100
millisecond
visual
region
this
and
that
and
the
shorter
term
memory
and
okay
we're
at
an
exhibit
so
there's
the
context
window
of

what
we've
seen
from
the
exhibition
and
then
it
even
expands
out
to
let's
see
how
slow
it
gets
in
the
very
slow
!
transformation
60
plus
seconds
but
then
the
transformative
capacity
and
Bert
talked
about
in

the
symposium
with
the
need
for
longer
term
lower
resolution
estimates
and
how
a key
engineering
question
right
now
is
being
able
to
adjust
basically
the
computational
resources
allocated
to
different
kinds
of
estimates
so
I
gave
an
example

of
if
you're
going
to
drive
three
quarters
of
the
way
around
a
roundabout
you
don't
need
to
know
what
centimeter
it's
going
to
be
at
you
just
want
to
know
what
lane
you're
going
to
be
in

but
the
next
centimeter
might
matter
for
the
next
centimeter
and
so
being
able
to
dynamically
tune
or
develop
hierarchical
structures
or
other
kinds
we're
using
our
CPU
cycles
and
other
resources
but
ultimately
attention
and
trust
on

like
we're
not
calculating
12
decimal
points
for
something
that
could
have
just
been
left
at
one
and
vice
versa
where
does
reasoning
sit
in
five
five
i think
it
depends
what
you
define
reasoning
as
but
it
depends

on
what
is
meant
by
reasoning
is
it
any
kind
of
reckoning
or
is
it
specifically
like
a
consciously
aware
of
one
but
that's
a
good
question
is
FEP
applicable
to
any
system
let's
vote
perennial
question
okay

i'm
going to
flip
a
coin
i'm
going to
flip
a
coin
this
is
going

be
a
real
coin
flip
i'm
going
to
use
the
real
entropy
of
my
room
it
was
not a
coin
with
heads
or
tails
so

it
was
not
a
conclusive
answer
but
i hope
that
in
future
!
figure out
more
general
generalist
tools
to
interact
with
bayesian
mechanics
i think
i remember
this question
being asked
and who
asked it
and the
mechanisms in
the brain
at the
on this topic
actually
the
first
presentation
in the

symposium
by
Andre
Bastos
this
was
very
much
like
like
welcome
to the
neuroscience
conference
like
not
not
not
grandstanding
at all
with a
theoretical
connection
just like
this is
how it's
being
visually
processed
here's
how we're
doing more
and more
measurements
here's
how
theory is
guiding

our
experiments
and it
did come
up to
this
like
multi-band
alpha
beta
and then
the
gamma
like
the
differential
which we
talked
about
I think
in one
or two
times
ago
differential
frequencies
well
cohort
three
sorry
Daniel
there was
this
question
I think
the
one
down

the
last
one
you
talked
about
about
is there
any
alternative
representation
for
something
like
that
yeah
are there
alternative
visualizations
that better
so I
think
we have
an additional
answer to
this question
now
because of
Chantal's
work
in string
diagrams
so
yeah
yeah
I
believe
Jana

made
this
image
and
then
I
know
that
Bert
has
done
a ton
of
work
in
alternative
visual
and
tactile
metaphors
and
then
on
the
more
formal
side
you're
absolutely
right
that
the
strain
diagram
may
even
or
some

of
the
representations
that
Sanjeev
is
developing
in the
fundamentals
work
certainly
we're seeing
things that
are like
familiar
but also
different
so
great works
cohort
four
that's
our
final
recorded
discussion
next
week
we'll
have
feedback
and talk
a little
bit
then we'll
take a
few
weeks

off
and
get
the gang
back
together
for
cohort
four
part
two
and
for
cohort
five
and
we'll
talk
more
next
week
about
the
timing
for
everything
and
see
what
works
for
people
so
thank
you
all
until
next

time

thanks

everyone

bye

you